

Shan Tirmizi

Full stack Engineer – London, United Kingdom

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 5+ years of experience specialising in backend systems, API performance, and scalable architectures. Proven track record of leading framework migrations, improving tail latency, and delivering production-grade features, including LLM-powered analysis tools and customer feedback pipelines. Strong frontend experience with React and Next.js, and effective in client-facing roles requiring clear technical communication.

TECHNICAL SKILLS

Languages: JavaScript (ES6), Python, TypeScript, Ruby

AI / LLM: Anthropic Claude API, LLM app integration, streaming, prompt engineering, RAG & agentic workflows

Backend Development: Node.js, Express, Python FastAPI, Flask, Ruby on Rails, Convex

Frontend Development: React.js, Next.js, HTML5, CSS3, Material ui, Shadcn ui, Tailwind

Mobile: React Native, Expo

Databases: MySQL, PostgreSQL, SQLite

Testing: Jest, Pytest, React Testing Library, Vitest

Methodologies: Agile, Scrum, CI/CD

Soft Skills: Strong Communication, Teamwork, Accountability, Empathy, Agile, Leadership

PROFESSIONAL EXPERIENCE

Full Stack Software Engineer

PolyAI, London, UK

June 2025 - Feb 2026

- Built and shipped Agent Analysis, an **LLM-powered** call analysis workflow enabling managers to evaluate a project's call performance using configurable categories, filtered sampling (random or manual selection), and actionable insights to optimise configuration
- Led a migration of a core backend service from **Flask to FastAPI**, reducing **p95 latency** by **~25–30%** on migrated endpoints and improving overall API responsiveness.
- Scoped and standardised a post-call Customer Satisfaction (CSAT) survey system, consolidating fragmented implementations into a single configurable pipeline and enabling consistent measurement of survey coverage and call satisfaction across projects.
- **Technologies used:** FastApi, Flask, Python, SQLAlchemy, PostgreSQL, TypeScript, Next.js, React, TanStack Query, Datadog

Full Stack Software Engineer

QuantSpark, London, UK

April 2024 - May 2025

- Enhanced existing **React** and **Flask** applications by implementing new features and modernising component architecture, resulting in **30% improved development velocity**
- Incrementally **migrated** legacy **JavaScript** components to **TypeScript** while ensuring all new features were developed using TypeScript, resulting in **40% fewer runtime errors** across converted modules
- Successfully onboarded new team members by providing in-depth codebase walkthroughs and

- documentation, significantly reducing onboarding time
- Regularly **engaged** with **clients** to gather and refine **project requirements**, effectively **communicating technical solutions** in a non-technical manner, fostering trust and ensuring **alignment** with **business objectives**
- Actively **participated** in **project planning** sessions, contributing valuable technical insights that shaped feature implementation and **improved project outcomes**
- **Technologies used**: React, Flask, TypeScript, JavaScript, Python, SQLAlchemy, MySQL, Tanstack Query, Jotai

Full Stack Software Engineer

Global Water Intelligence (GWI), Oxford, UK

September 2022 - April 2024

- Successfully **rewrote** the UPM codebase from **Ruby on Rails** to **Next.js**, resulting in a **50% increase** in application performance, **30% reduction** in server response times, and enhanced scalability to accommodate future growth
- Implemented and maintained the **company's first documentation** system, reducing new developer **onboarding time** by **81%**, from **1.5 weeks** to just **2 days**
- Made code reviews mandatory and actively participated, providing feedback that led to a 30% improvement in overall code quality and 20% reduction in production issues
- **Led** the **development** of 'GWI Opportunity Exchange', the **first community-sourced marketplace** for the **water sector business**, using React, Redux, MySQL, and Ruby on Rails, resulting in a **15% increase** in **user engagement** within the first six months
- Contributed to the development of the UPM University website, designed to provide online courses for the microelectronics industry, leveraging Next.js, Node.js, Express, and Sanity
- Balanced new software development with improvements to legacy projects, demonstrating versatility
- **Technologies used**: Next.js, React, Redux, MySQL, Ruby on Rails, Node.js, Express, Sanity

Frontend Developer

InvestCloud, London, UK

May 2021 - August 2022

- Promoted to lead the **£9.4 million** project, where I implemented advanced communication techniques, **achieving 20% reduction** in development time and **25% improvement** in team coordination
- Successfully led a project from start to finish, delivering it two weeks ahead of schedule, thereby **increasing client satisfaction** by **25%**
- Addressed and **fixed 95% of critical bugs** in a major project within **three months**, enhancing system stability and resulting in **25% reduction** in **customer support tickets**
- Developed and implemented custom Screens & **Dashboards** for **Financial Firms**, improving **data visualisation** and user experience, leading to **35% increase** in **client engagement**
- **Collaborated** directly with **clients** and **stakeholders** to understand their needs, translating **technical requirements** into actionable plans and ensuring clear communication of project progress, which contributed to an increase in **client satisfaction**
- **Interviewed, trained** and **supervised** new employees
- Effectively scoped project requirements using Agile & Scrum principles, resulting in 20% reduction in turnaround time for project deliverables
- **Technologies used**: React.js, JavaScript, TypeScript

Technical Projects

Visa Atlas

Personal Project · Cross-platform iOS & Android travel app

Mar 2026 - Present

- Built an **AI itinerary generator** that produces complete, structured trip plans (flights, lodging, dining, activities) with **streaming Claude responses** for real-time output.
- Engineered an agentic, **web-grounded day-planner** that turns a start point, transport mode and vibe into a **geocoded, routed** day itinerary.
- Shipped a **Claude-powered visa assistant** delivering grounded, passport-aware answers backed by a structured visa-rules dataset.
- Architected a **serverless realtime backend** on **Convex** (29 modules: queries, mutations, scheduled actions, auth) with a streaming **Anthropic proxy** and per-user rate limiting.
- Built an interactive **MapLibre** world map and real-time **trip collaboration** (revocable share links, voting), with Apple and email-OTP auth and push notifications.
- **Technologies used:** Expo, React Native, TypeScript, Convex, Anthropic Claude, MapLibre, Reanimated, Jest

HabitQuest

Personal Project · Gamified habit-tracking mobile app

Feb 2026 - Apr 2026

- Built “**Dr. Sage**”, a **Claude-powered AI coach** that analyses habits, journal entries and streaks to deliver personalised insights and conversational guidance.
- Designed an **AI memory system** that extracts and stores user preferences, blockers and strategies over time, so coaching improves with continued use.
- Added **AI-generated daily challenges** and an AI goal-planning assistant, all powered by Claude.
- Built a full **RPG gamification** layer — XP and levelling, HP, evolving companions, weekly bosses and a skill tree — to drive habit retention.
- Architected a **Convex** realtime backend (**19 tables**, 25 modules) with multi-provider auth (email, Google, Apple), push notifications and i18n.
- **Technologies used:** Expo, React Native, TypeScript, Convex, Anthropic Claude, Reanimated, PostHog

Awards & Achievement

Hackathon (Le Wagon 2021 Winner)

Mar 2021

- Had **24 hours** to create a **full-stack app** for a beverage company.
- Successfully created the app and **won** the hackathon against **92 candidates**.

Education

Le Wagon Coding Bootcamp (Web Development)

Oct 2020 - Dec 2020

- Le Wagon is a 9-week intensive Bootcamp that focuses on Ruby on Rails, Ruby, CSS, Bootstrap, JavaScript, SQL, PostgreSQL, MVC Architecture, Object-Oriented Programming, Git, GitHub and HTML.
- Built two full-stack projects remotely using **Agile Principles**.

Strode's College (A-levels)

Sept 2015 - Jun 2017

- Physics (A), Mathematics (A) and Economics (A).